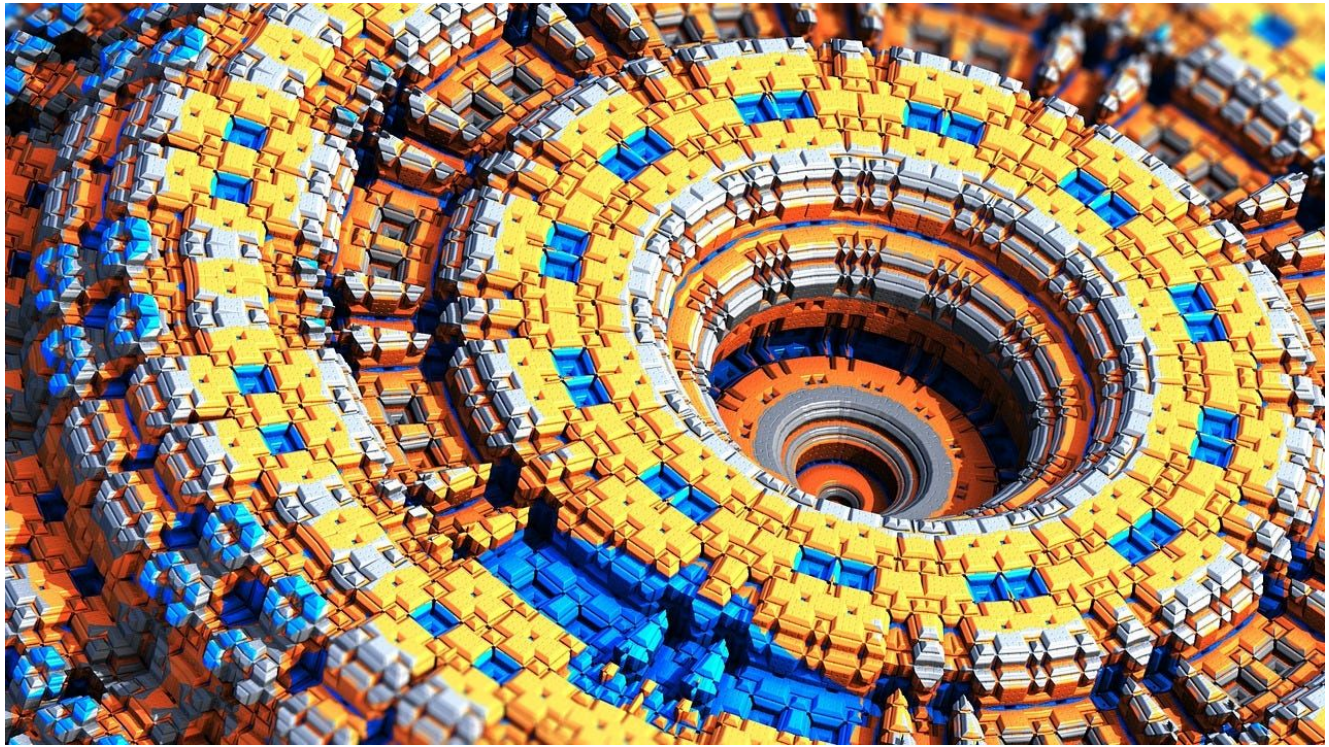
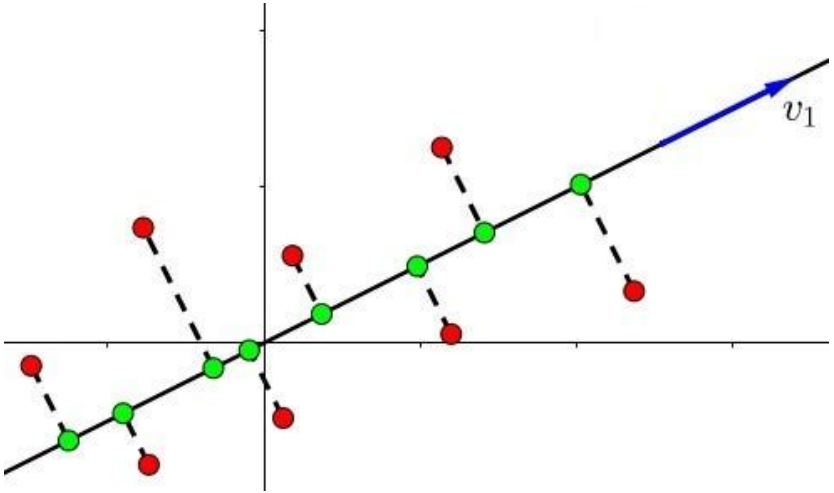
Dimensionality reduction

Linear methods

Motivation



Linear transformations - today



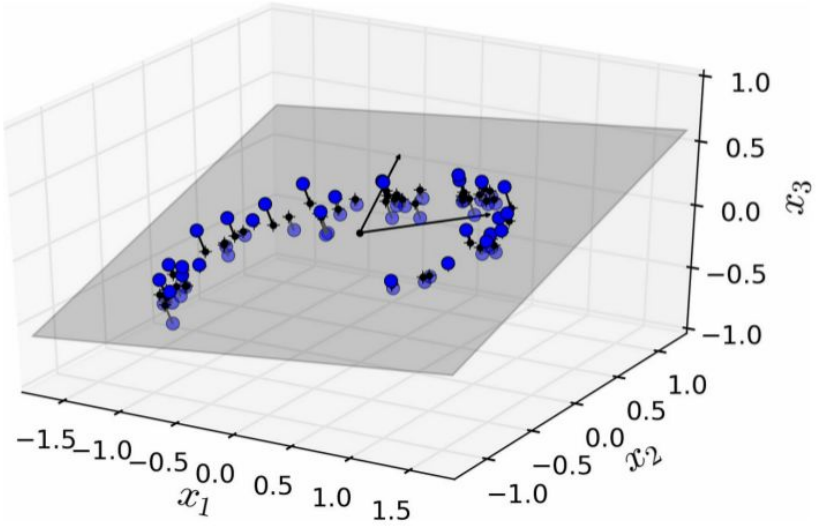
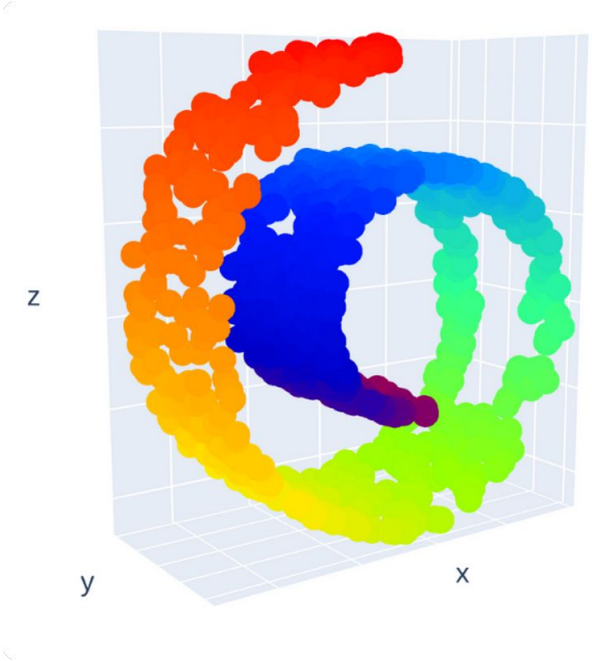
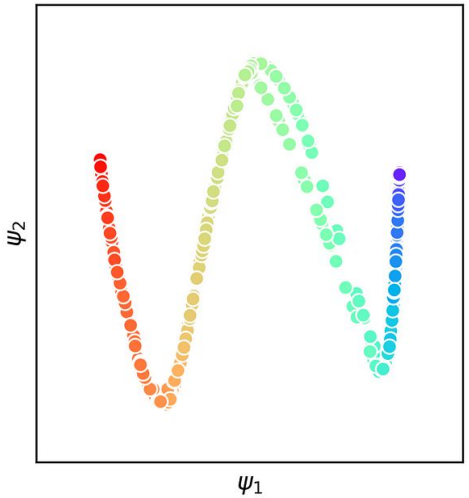
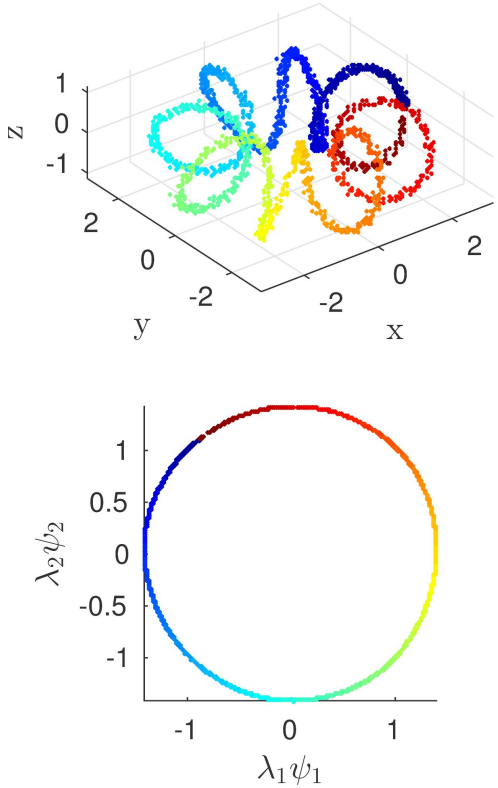


Figure 8-2. A 3D dataset lying close to a 2D subspace

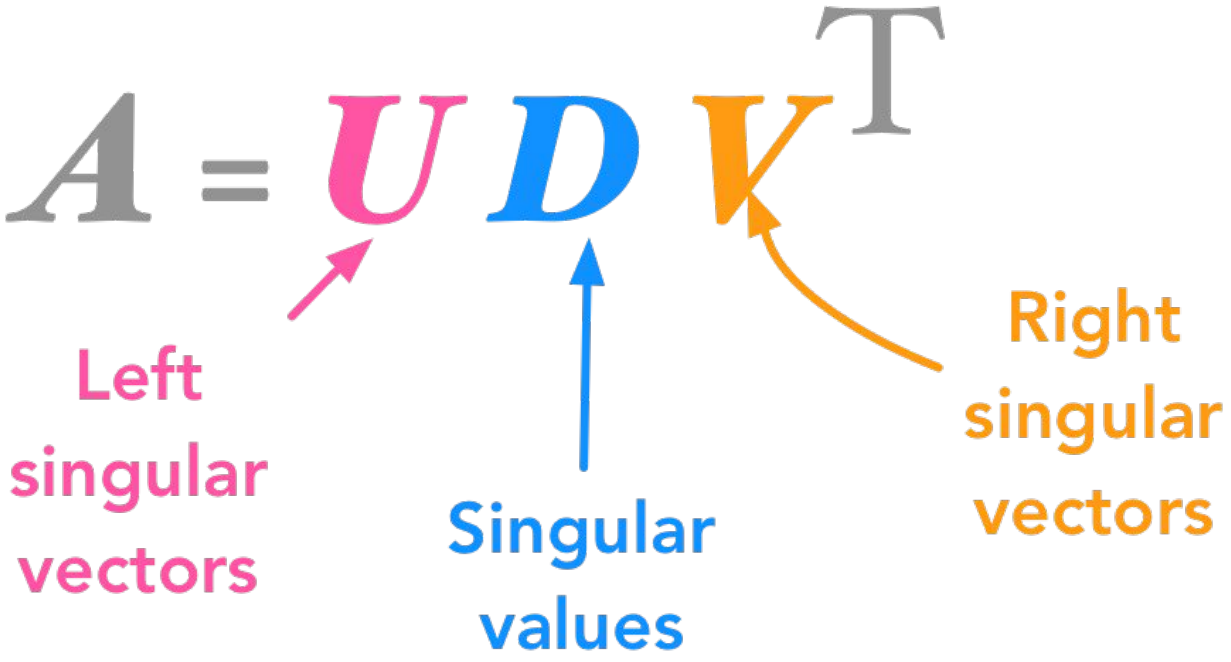
Nonlinear transformations - some other time



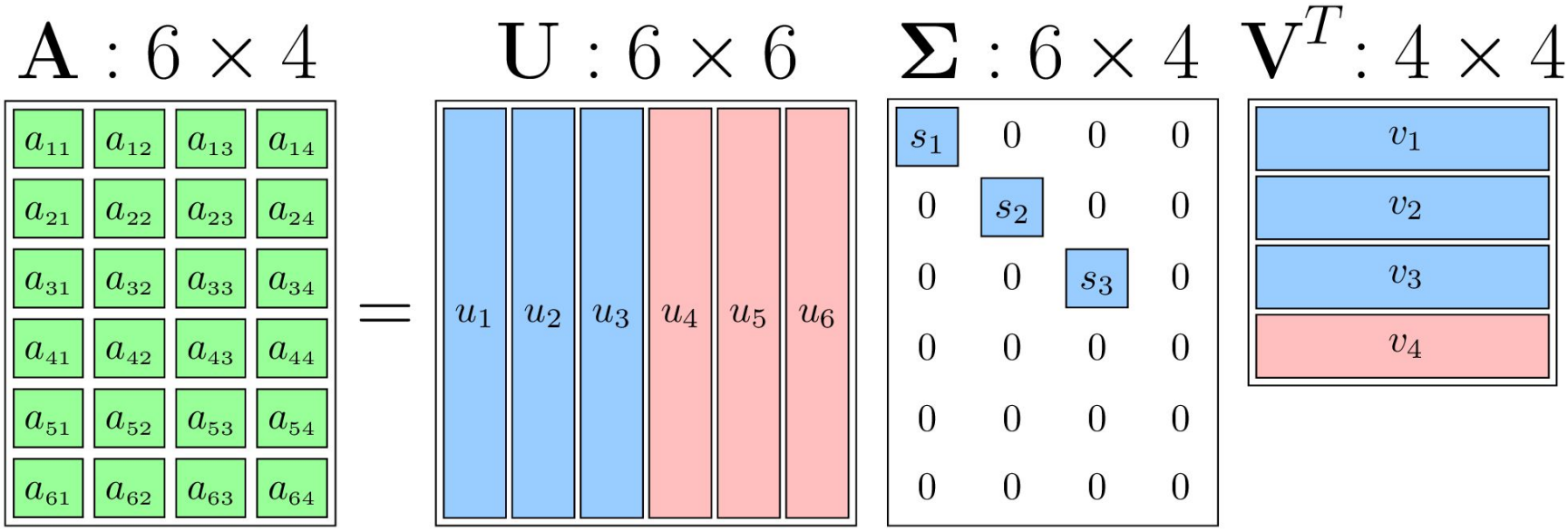




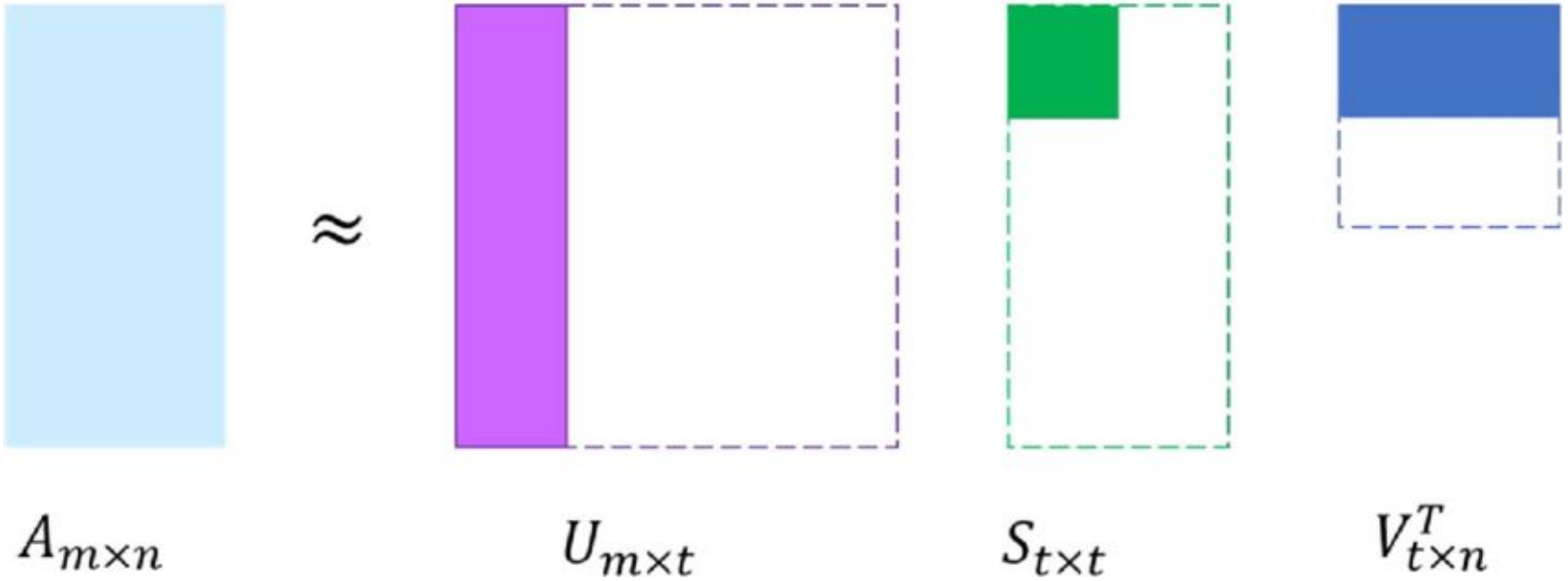
Singular value decomposition (SVD)



SVD


$\mathbf{A} : 6 \times 4$
$\mathbf{U} : 6 \times 6$
$\mathbf{\Sigma} : 6 \times 4$
$\mathbf{V}^T : 4 \times 4$
a_{11} a_{12} a_{13} a_{14}
a_{21} a_{22} a_{23} a_{24}
a_{31} a_{32} a_{33} a_{34}
a_{41} a_{42} a_{43} a_{44}
a_{51} a_{52} a_{53} a_{54}
a_{61} a_{62} a_{63} a_{64}
=
u_1 u_2 u_3 u_4 u_5 u_6
s_1 0 0 0
0 s_2 0 0
0 0 s_3 0
0 0 0 0
0 0 0 0
0 0 0 0
v_1
v_2
v_3
v_4

Truncated SVD



The Eckart – Young Theorem (1936)

says that the "truncated SVD" matrix

$$A_k = U_k D_k V_k^T$$

solves the least norm problem

minimize $F(B) = \|A - B\|_F^2$

subject to $\text{rank}(B) \leq k$

(Also called the Schmidt-Mirsky Theorem.)

Image compression using SVD



Image compression using SVD

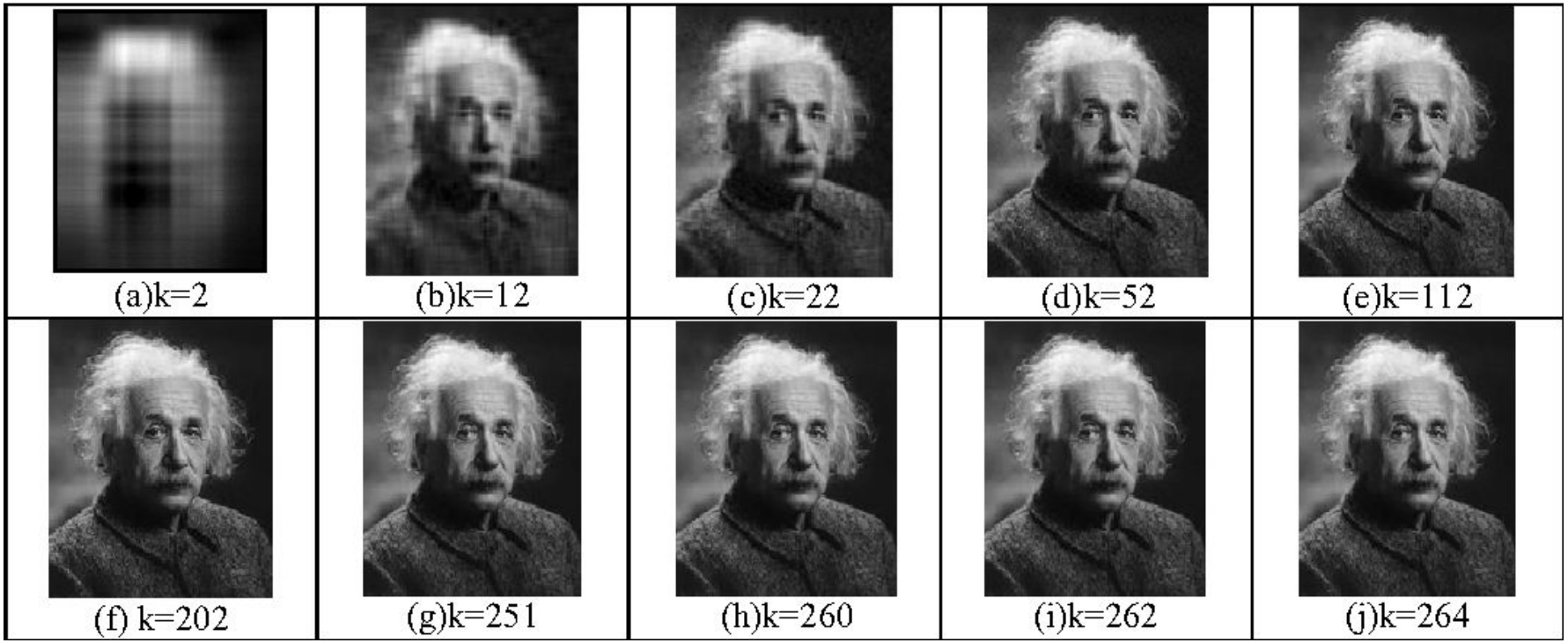
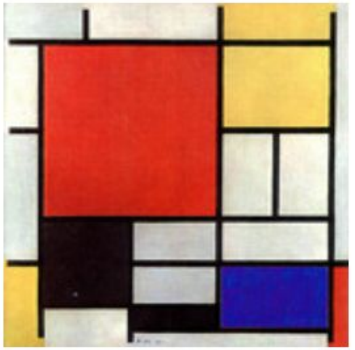


Image compression using SVD

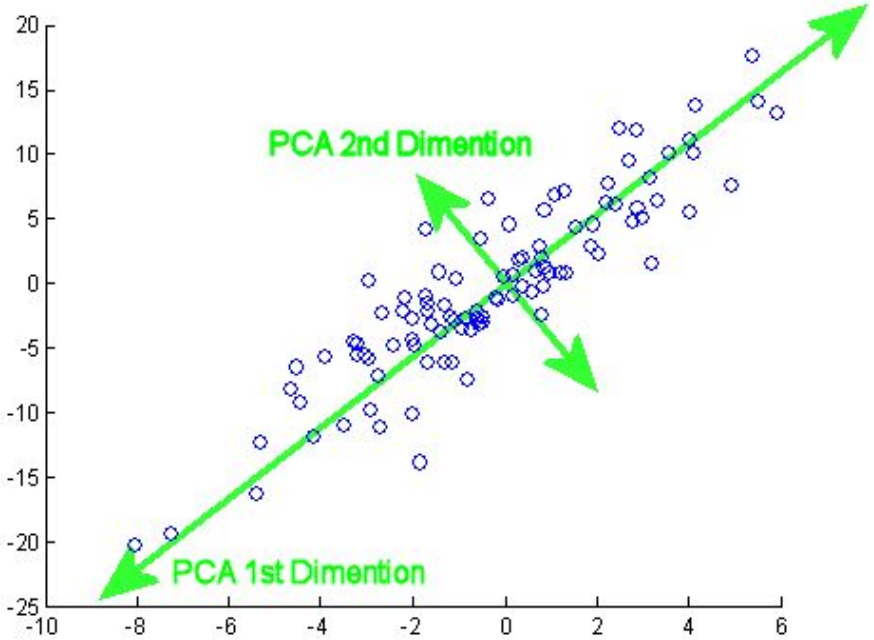
Can we do the same with color pictures? DEMO

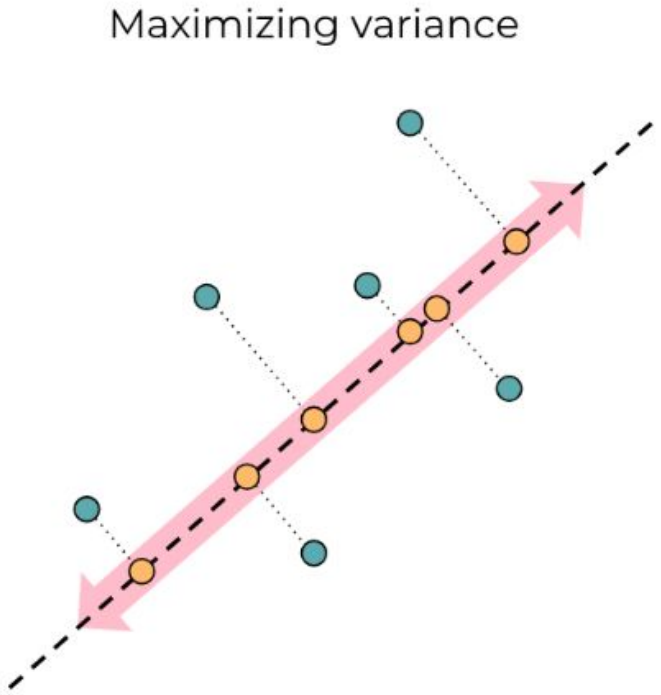




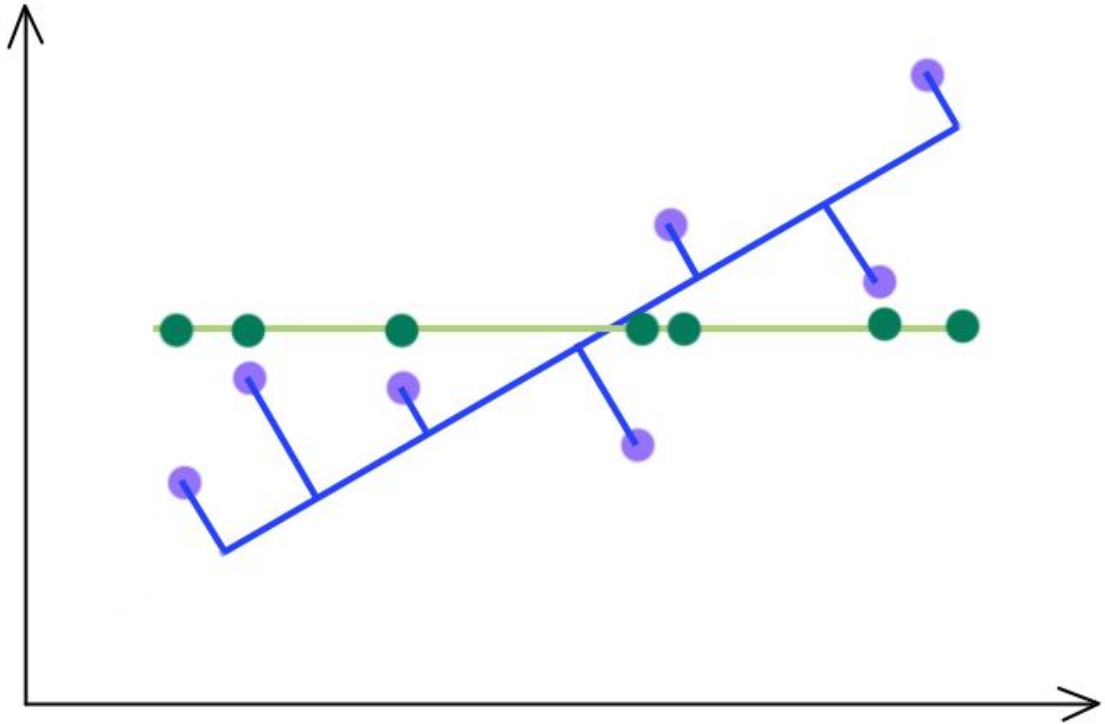


Principal components analysis (PCA)



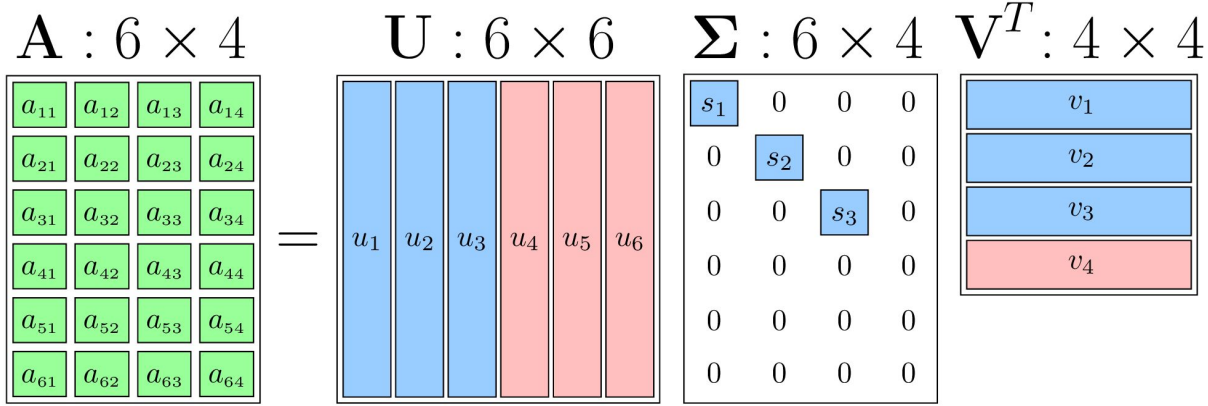


PCA: which component is principal?



PCA: Calculation using SVD

Suppose there are n data points $\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_n \in \mathbb{R}^d$, arranged as the rows of the matrix $\boldsymbol{A} \in \mathbb{R}^{n \times d}$. Now regard $\boldsymbol{X}$ as a random vector with the uniform distribution on the n data points. Then $\mathbb{E}(\boldsymbol{X}\boldsymbol{X}^\top) = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{a}_i \boldsymbol{a}_i^\top = \frac{1}{n}\boldsymbol{A}^\top \boldsymbol{A}$: top-$k$ eigenvectors of $\frac{1}{n}\boldsymbol{A}^\top \boldsymbol{A}$ are top-k right singular vectors of $\boldsymbol{A}$. Hence, rank-k uncentered PCA (as in Theorem 5.4) corresponds to the subspace spanned by the top-k right singular vectors of $\boldsymbol{A}$.



Digits classification

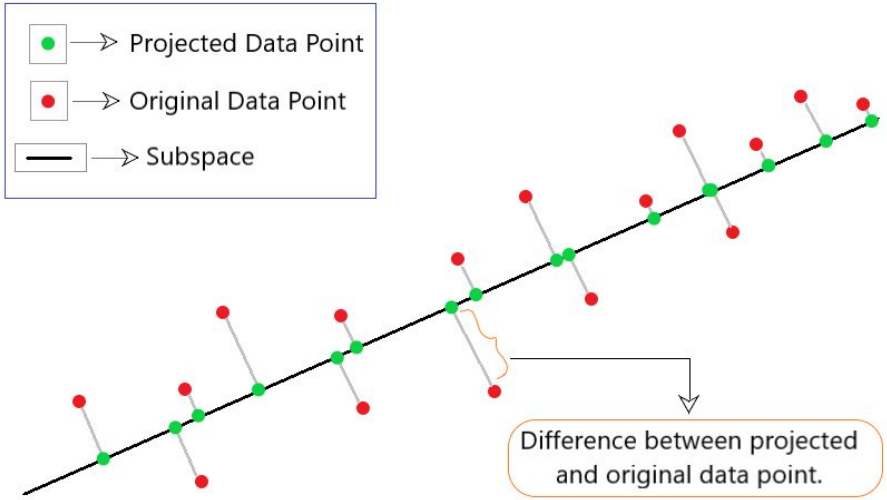
Ideas:

- clustering?
- **“learning” subspaces?**
- neural networks?



Best-fit subspaces

$$e(\mathcal{W}) = \sum_{i=1}^{n} \|\mathbf{a}_i - \mathbf{P}_{\mathcal{W}}\mathbf{a}_i\|_2^2$$



Best-fit subspaces

$$e(\mathcal{W}) = \sum_{i=1}^{n} \|\mathbf{a}_i - \mathbf{P}_{\mathcal{W}}\mathbf{a}_i\|_2^2 = \sum_{i=1}^{n} \|\mathbf{a}_i - WW^T\mathbf{a}_i\|_2^2 = \|A - WW^TA\|_F^2$$

$$\|A - WW^TA\|_F \longrightarrow \min_{W, W^TW = I_k}$$

The Eckart – Young Theorem (1936)

says that the "truncated SVD" matrix

$$A_k = U_k D_k V_k^T$$

solves the least norm problem

minimize $F(B) = \|A - B\|_F^2$

subject to $\text{rank}(B) \leq k$

(Also called the Schmidt-Mirsky Theorem.)

Best-fit subspaces

$$\|A - WW^T A\|_F \geq \|A - U_k \Sigma_k V_k^T\|_F = \|A - U_k U_k^T A\|_F$$

because

$$U_k U_k^T U \Sigma V^T = U_k U_k^T \begin{pmatrix} U_k & U_k^\perp \end{pmatrix} \Sigma V^T = U_k \begin{pmatrix} I_k & \mathbf{0} \end{pmatrix} \Sigma V^T = U_k \Sigma_k V_k^T$$

Digits classification with best-fit subspaces

$$z_i = U^{(i)}(U^{(i)})^T z$$

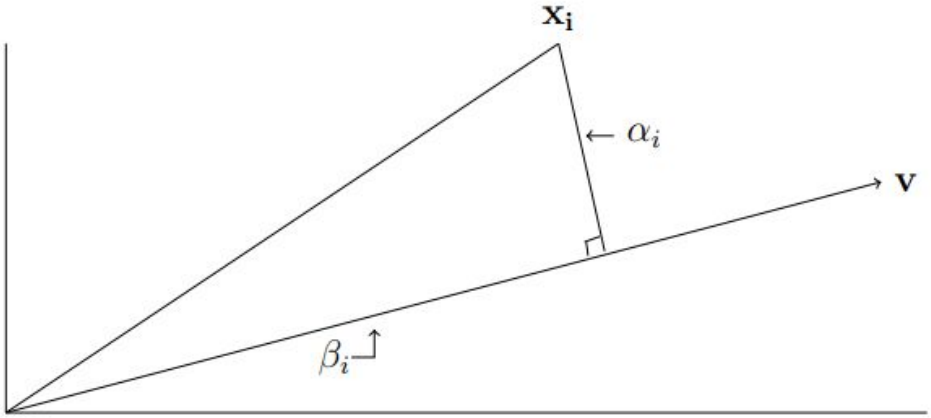
$$j = \arg\min_i \|z - z_i\|_2$$

We learned

- SVD
- PCA
- Best-fit subspaces

But, PCA and “best-fit subspaces” are actually the same thing

Connection between PCA and best-fit subspaces

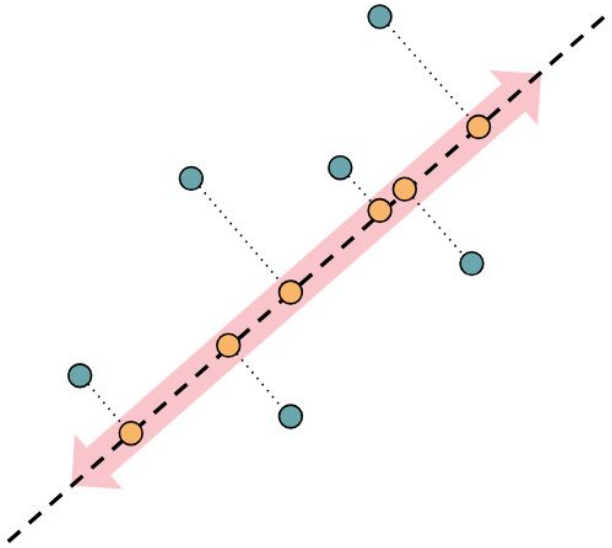


Minimizing $\sum_i \alpha_i^2$ is equivalent to maximizing $\sum_i \beta_i^2$

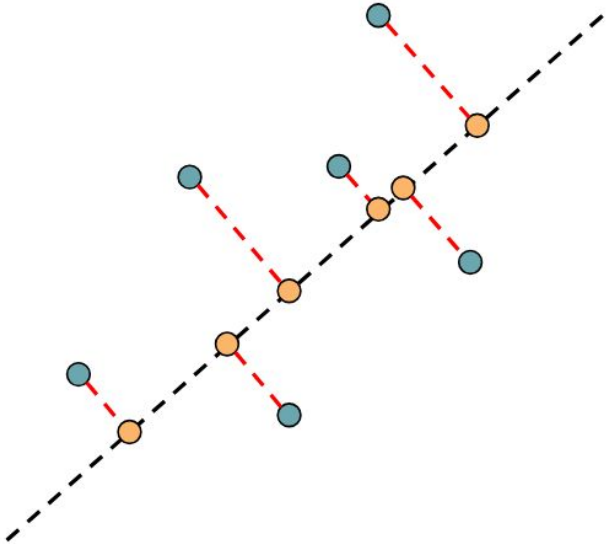
Figure 3.1: The projection of the point $\mathbf{x_i}$ onto the line through the origin in the direction of $\mathbf{v}$.

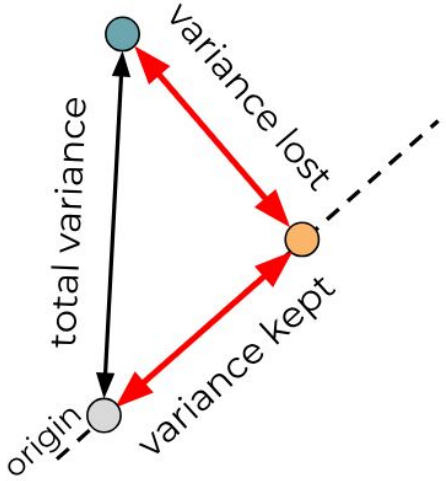
Connection between PCA and best-fit subspaces

Maximizing variance



Minimizing
reconstruction error





Thank you!

Connection between PCA and learning subspaces

Project a point $\mathbf{a_i} = (a_{i1}, a_{i2}, \ldots, a_{id})$ onto a line through the origin. Then

$$a_{i1}^2 + a_{i2}^2 + \cdots + a_{id}^2 = (\text{length of projection})^2 + (\text{distance of point to line})^2 .$$

See Figure 3.1. Thus

$$(\text{distance of point to line})^2 = a_{i1}^2 + a_{i2}^2 + \cdots + a_{id}^2 - (\text{length of projection})^2 .$$

Since $\sum_{i=1}^{n} (a_{i1}^2 + a_{i2}^2 + \cdots + a_{id}^2)$ is a constant independent of the line, minimizing the sum of the squares of the distances to the line is equivalent to maximizing the sum of the squares of the lengths of the projections onto the line. Similarly for best-fit subspaces, maximizing the sum of the squared lengths of the projections onto the subspace minimizes the sum of squared distances to the subspace.